

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0089

Mcross 138 kV
November 16, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0089. The Interconnection Customer proposes a 534.7 MW large load (data center) facility interconnecting at the Mcross 138 kV substation in the West Virginia area. The request was received on December 27, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 79 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$126,936,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Mogadore to Mcross 138 KV: 304.1% loading (181 MVA) following the outage of the Wastewater to Scarbro 138 KV; DFAX = 100.0%.

Wastewater to Mcross 138 KV: 270.9% loading (199 MVA) following the outage of the Mogadore to Mcross Substation 138 KV; DFAX = 100.0%.

For the loss of Mcross to Mogadore 138 KV, the Wastewater to Scarbro 138 KV line loads to 253.8% of its 217 MVA rating. Project DFAX: 100.0%.

The Sidney Term to Meadow Bridge 138 KV is overloaded to 205.6% of its 187 MVA emergency rating for the loss of the Scarbro to Wastewater 138 KV. The project contributes 69.8% DFAX (373.1 MW) to this overload.

Scarbro to North Sweetwater 138 KV: 198.2% loading (253 MVA) following the outage of the Mogadore to Mcross 138 KV; DFAX = 78.2%.

For the loss of Wastewater to Scarbro 138 KV, the Sidney erm to Mogadore 138 KV loads to 196.5% of its 195 MVA rating. Project DFAX: 69.8%.

For the loss of Cabin Creek to Hawks Nest Hydro 345 KV line, the Hinton Ap to Antelope Vly Sta 138 KV Line loads to 162.6% of its 209 MVA rating. Project DFAX: 34.4%.

For the loss of Wastewater to Scarbro 138 KV, the Meadow Bridge to Antelope Vly Station 138 KV loads to 159.5% of its 182 MVA rating. Project DFAX: 48.2%.

Bisti to N Sweetwater 138 KV: 135.8% loading (244 MVA) following the outage of the Mogadore to Mcross Substation 138 KV L; DFAX = 48.7%.

Stone orral to Bisti 138 KV: 133.4% loading (254 MVA) following the outage of the Mogadore to Mcrss 138 KV; DFAX = 48.7%.

For the loss of Antelope Vly Sta to Meadow Bridge 138 KV, the Heyburn to Frio Town 138 KV loads to 117.3% of its 144 MVA rating. Project DFAX: 30.2%.

The Shingle Springs to Hawks Nest Hydro 138 KV is overloaded to 110.6% of its 337 MVA emergency rating for the loss of the Hawks Nest Hydro to Vergennes 9 Substation 345 KV. The project contributes 25.5% DFAX (136.6 MW)

to this overload.

For the loss of Vergennes 9 to Hawks Nest Hydro 345 KV, the Stone Corral to Shingle Springs 138 KV loads to 109.3% of its 296 MVA rating. Project DFAX: 22.6%.

The Vergennes 9 345-138 kV Auto is overloaded to 106.2% of its 384 MVA emergency rating for the loss of the Hawks Nest Hydro 345-138 kV TX. The project contributes 20.3% DFAX (108.8 MW) to this overload.

The Rio Grande to Mogdore 138 KV is overloaded to 104.1% of its 191 MVA emergency rating for the loss of the Sidney Term to Meadow Bridge 138 KV. The project contributes 34.9% DFAX (186.6 MW) to this overload.

Phelan Substation to Stone Corral 138 KV: 102.2% loading (253 MVA) following the outage of the Stone Corral to Bisti 138 KV; DFAX = 19.8%.

Stone Corral to Keweenaw 138 KV line: 101.3% loading (211 MVA) following the outage of the Hawks Nest Hydro 345-138 kV XFMR; DFAX = 10.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mogadore to Mcross 138 KV	Rebuild line with higher-capacity conductor	\$5,970,000	100% (\$5,970,000)
Wastewater to Mcross 138 KV	Rebuild line with higher-capacity conductor	\$18,963,000	100% (\$18,963,000)
Wastewater to Scarbro Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$5,393,000	100% (\$5,393,000)
Sidney Term to Meadow Bridge 138 KV	Rebuild line with higher-capacity conductor	\$4,377,000	100% (\$4,377,000)
Scarbro to North Sweetwater 138 KV	Rebuild line with higher-capacity conductor	\$2,808,000	100% (\$2,808,000)
Sidney Term to Mogadore 138 KV	Rebuild line with higher-capacity conductor	\$3,430,000	100% (\$3,430,000)
Hinton Ap to Antelope Vly Sta 138 KV Line	Rebuild line with higher-capacity conductor	\$14,915,000	100% (\$14,915,000)
Meadow Bridge Substation to Antelope Vly Sta 138 KV	Rebuild line with higher-capacity conductor	\$22,790,000	100% (\$22,790,000)
Bisti to N Sweetwater 138 KV	Rebuild line with higher-capacity conductor	\$6,633,000	100% (\$6,633,000)
Stone orral to Bisti 138 KV	Rebuild line with higher-capacity conductor	\$1,894,000	100% (\$1,894,000)
Frio Town to Heyburn 138 KV	Rebuild line with higher-capacity conductor	\$9,939,000	100% (\$9,939,000)
Shingle Springs to Hawks Nest Hydro 138 KV	Rebuild line with higher-capacity conductor	\$2,312,000	100% (\$2,312,000)
Shingle Springs to Stone Corral 138 KV	Reconductor line	\$1,159,000	100% (\$1,159,000)
Vergennes 9 Substation 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$18,358,000	100% (\$18,358,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rio Grande Substation to Mogadore Substation 138 KV	Reconductor line	\$1,206,000	100% (\$1,206,000)
Phelan Substation to Stone Corral 138 KV	Reconductor line	\$1,150,000	100% (\$1,150,000)
Stone Corral to Keweenaw 138 KV line	Reconductor line	\$5,639,000	100% (\$5,639,000)
	Total allocated to project		\$126,936,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.